
PURPOSE

This is the foundational astronomical model for MK147 during the Blackstone University era.

The goal is NOT hard science realism.

The goal is:

Plausible
+
Stable
+
Readable
+
Consistent

The reader should experience:

Two Suns
Beautiful Skies
Strange Celestial Events
Long Days
Stable Seasons

without requiring a physics degree.

THE STAR SYSTEM

System Type:

Circumbinary System

Two stars orbit one another.

The planets orbit both stars together.

This allows for:

- Stable planetary orbits
- Stable agriculture
- Stable civilization
- Stable universities
- Stable kingdoms

while preserving:

- Twin sunrises
- Twin sunsets
- Two shadows
- Binary sky phenomena

THE STARS

Star A

~0.55 Solar Mass

Star B

~0.45 Solar Mass

Combined System

~1 Solar Mass Equivalent

This permits:

Earth-like climate behavior

without requiring:

Earth-like skies

THE PLANETS

Planet I

"The Ash World"

Inner world

Extremely hot

Mercury/Venus analog

Hostile

Little civilization

Minimal importance to story

Planet II

"Kira's World"

Primary setting

Goldilocks World

Temperate

Oceans

Forests

Agriculture

Kingdoms

Universities

Blackstone University

All major story events occur here.

Planet III

"The Frontier World"

Cold

Sparse

Difficult

Exploration focused

Mars-like role

Potential future stories

Potential colonies

Potential research stations

Planet IV

"The Great Giant"

Gas giant

Ice giant

Ring system possible

Bright night-sky object

Astronomically significant

Frequently visible from Kira's world

THE GOLDBLOCKS ZONE

Planet II occupies the habitable zone.

Conditions permit:

- Liquid water
- Stable weather
- Agriculture
- Large ecosystems
- Human civilization

The world intentionally remains:

Familiar enough to write

Different enough to feel unique

THE SKY

The sky is one of the defining features of the world.

Common observations:

Twin Sunrises

Occasionally both stars rise visibly.

Rarely identical.

One generally dominates.

Twin Sunsets

One of the most beautiful events on the planet.

Often referenced in poetry.

Common location for:

- Romantic scenes
- Reflection scenes
- Major story moments

Dual Shadows

Depending on season and geometry.

A person may cast:

Two overlapping shadows.

Long Twilight

Certain regions experience unusually long
sunrise and sunset periods.

Creates striking visual effects.

CELESTIAL EVENTS

Every few years:

Rare celestial alignments occur.

Examples:

- Double eclipses
- Twin-star conjunctions
- Bright giant-planet events
- Ring-planet visibility events

These events become:

Festivals

Religious observances

Scientific opportunities

Tourism events

University research opportunities

THE DAY

Current Canon:

31 Hour Day

The planet rotates more slowly than Earth.

The longer day affects:

- Agriculture
- Work schedules
- Education
- Travel
- Social customs

The inhabitants do not perceive this as unusual.

To them:

31 Hours

=

Normal

CULTURAL CONSEQUENCES

Longer days create:

Longer Study Periods

Universities can schedule fewer but longer classes.

Matches Blackstone's educational model.

Longer Travel Windows

Travel becomes easier.

More productive daylight.

More practical exploration.

Longer Work Cycles

Certain professions operate for extended periods.

Especially:

- Adventurers
- Researchers
- Healers
- Farmers

THE CALENDAR

UNDEFINED

Current Status:

To Be Developed

Known Requirements:

- Must support 31-hour days
- Must support four seasons
- Must support educational semesters
- Must support agriculture
- Must support festivals

Future work:

Weeks

Months

Years

Seasonal cycles

BLACKSTONE IMPLICATIONS

The university exists within this astronomical system.

Consequences:

- Night study periods become longer.
- Celestial events become university-wide events.
- Astronomy becomes a legitimate field.
- Quantum Field Studies gains additional relevance.
- Natural Philosophy gains additional relevance.

Students commonly observe:

Twin sunsets from Blackstone courtyards.

AUTHOR NOTE

The purpose of the star system is not to create a hard-science setting.

The purpose is to create:

A believable sky.

A believable calendar.

A believable world.

A believable civilization.

Such that all higher-order structures may inherit from it naturally:

Planet
↓
Civilization
↓
Kingdom
↓
City
↓
University
↓
Student
↓
Story

This star system serves as the astrophysical foundation upon which Blackstone University, the Kingdom, Kira, and all future MK147 stories are built.